

TOKENS

Token is the smallest unit of the Java programming language, just like English language has alphabets. The symbols we use in Java like

- dot/access operator(.)
- comma/separator operator(,)
- semi-colon/terminator(;)
- open of block({)
- end of block(})

are all examples of tokens.

But we will focus on the most important three tokens in Java. They are:

1. Keywords
2. Identifiers
3. Literals

KEYWORDS:

They are **pre-defined, compiler aware, reserved words** in Java. Their meaning and purpose cannot be changed by programmers. They are always written in lower case. They are finite in number, around 50+ words.

Eg: class, public, static, void

IDENTIFIERS:

There are many components in Java like variables, methods, class, interface and enum. The names given to these components are called identifiers. They are user-defined words. When a programmer names a component of Java, rules and conventions have to be followed.

Eg: ClassName, methodName, variablename

Rules for Identifiers:

Rules are mandatory as they are checked by the compiler. If rules are not followed, it will throw Compile Time Error. The following are the four rules:

1. It should NOT be a keyword.
2. It should NOT start with a number.
3. It should NOT be special character except \$ and _.
4. It should NOT have space character.

\$ can be used alone but _ has to be used with some other characters.

Conventions for Identifiers:

Conventions are not mandatory but highly recommended. They are not checked by the compiler but they are standards which are adopted and followed by Java programmers worldwide.

Let us see the conventions to name three important components in Java.

Class Name:

It always starts with upper case and every word starts with upper case if it is a multi-word name. This way of naming is called Pascal casing or Upper Camel casing.

Eg: class RajaRamMohanRoy

```
{  
}
```

Method Name:

It always starts with lower case and every word from second word starts with upper case if it is a multi-word name. This way of naming is called Camel casing.

Eg: public static void rajaRamMohanRoy()

```
{  
}
```

Variable Name:

It is always written in lower case.

Eg: int rajarammohanroy = 10;

LITERALS:

The programmer may have to give data in the program directly in some cases. That data or values given by the programmer are called as Literals. There are four types of literals in Java. They are:

1. Numbers
2. Characters
3. Strings
4. Boolean

Number Literals:

Number literals constitute both positive and negative, whole numbers and decimal numbers along with zero.

Eg: 52, -85, 0, 7.65, -6.87

Character Literals:

Any single character enclosed in single quotes is called as character in Java. The length of character literal is always one. The length cannot be zero or more than one.

Eg: 'd', 'G', '5', ' ', '0'

String Literals:

Any data enclosed in double quotes are called as Strings in Java. The length of String is not defined. The length can be zero or any positive length. It accepts alphanumeric data also.

Eg: "", " ", "aed", "AWDTahdujol", "525gdtrh"

Boolean Literals:

The Boolean literals are used for logical data in Java. The Boolean representation of ON/logical HIGH/YES in Java is true. It has only two values, true and false.

YES/ON/logical HIGH: true

NO/OFF/logical LOW: false